

Deep Learning for Economics and Finance

Stochastic IAMs with DEQNs, GP surrogates, Sobol sensitivity indices, constrained
Pareto-improving carbon-tax design

Simon Scheidegger

HEC, University of Lausanne

Based on: Kübler, Scheidegger & Surbek (2026, forthcoming, *JPE: Macroeconomics*;
arXiv:2507.01704)

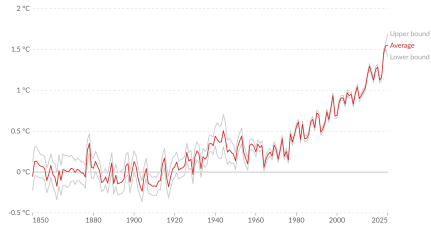
Companion code: [https://github.com/sischei/JPE_Macro_Using_ML_to_compute_constrained_
optimal_carbon_tax_rules](https://github.com/sischei/JPE_Macro_Using_ML_to_compute_constrained_optimal_carbon_tax_rules)

Materials: [https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_
Dynamic_Economic_Models](https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models)

The Challenge: Climate Risk & Heterogeneity

Annual temperature anomalies relative to the pre-industrial period, World

The difference in average land-sea surface temperature compared to the 1861-1890 mean, in degrees Celsius.



Data source: Met Office Hadley Centre - HadCRUT5 (2025)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Note: The period 1861-1890 is used as the baseline to measure temperature changes relative to pre-industrial times, [as recommended by the source](#).



- ▶ **Systemic Risk:** Climate change poses a fundamental threat to global economic stability and human welfare.
- ▶ **Non-Linear Dynamics:** Tipping points introduce the risk of irreversible, catastrophic environmental shifts.
- ▶ **Distributional Impact:** Effects will be highly heterogeneous, creating profound equity challenges across regions and generations.
- ▶ **Literature Context:** These macro-financial linkages are detailed in the recent review Fernandez-Villaverde, Gillingham, and Scheidegger (2025).

Motivation: Designing Optimal Policy

- ▶ **The Consensus:** Carbon pricing is widely regarded as the most efficient tool to mitigate emissions.
- ▶ **The Difficulty:** In a world with heterogeneous agents (e.g., OLG), simple taxes create winners and losers, threatening political feasibility.
- ▶ **The Objective:** We need a systematic way to compute **constrained optimal taxes** that:
 - ▶ Account for stochastic risks (tipping points).
 - ▶ Respect political constraints (Pareto improvements).

The Gap: Welfare Analysis in Stochastic OLG IAMs

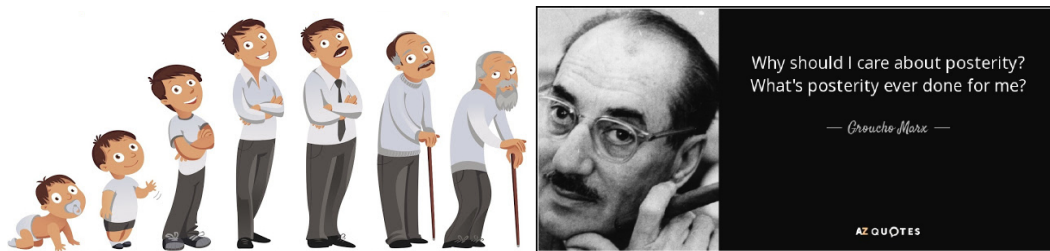


Figure: Overlapping Generations.

- ▶ **Current Limitations:** Most climate-macro research relies on [simplifying assumptions](#) (e.g., representative agents) or heuristic "guessing" to find Pareto improvements.
- ▶ **The Missing Tool:** A framework for [global solutions](#) in high-dimensional, stochastic environments that explicitly handles:
 - ▶ **Heterogeneity:** Winners vs. Losers.
 - ▶ **Uncertainty:** Tipping points and damage shocks.

Our Contribution: A Scalable Policy Framework

1. The Method

Step 1: DEQN

$$P(x_{\text{state},t}, \vartheta)$$

Step 2: GP(θ)

$$\widehat{\text{QoI}}(\vartheta)$$

Step 3: Optimizer

$$\arg \max \widehat{\text{QoI}}(\vartheta)$$

A 3-step ML pipeline to alleviate the [curse of dimensionality](#).

2. The Application

We compute the first [globally optimal, Pareto-improving carbon tax](#) in a stochastic OLG model with tipping points.

3. A Generic Blueprint

A universal framework for **almost any** stochastic heterogeneous-agent model. It allows researchers to both [solve global equilibria](#) and derive [constrained-optimal policies](#) in settings previously deemed intractable.

An Incomplete List of Related Literature

Climate Risk in OLG Models

Existing work studies intergenerational conflict and climate damages (Kotlikoff et al., 2021a; Weitzman, 2012; Karp and Rezai, 2014).

- + **Our Gap:** Most rely on linearization or ignore the *political constraint* of Pareto improvements under uncertainty (e.g., tipping points).

Optimal Ramsey Policy with Heterogeneity

Established frameworks for fiscal/monetary policy in HA models (Bhandari et al., 2013; Nuño and Thomas, 2020; Douenne et al., 2024).

- + **Our Gap:** We enforce *simple rules* (linear taxes) to make the policy implementable and interpretable.

Deep Learning for High-Dimensional Dynamic Stochastic Models

Using Neural Networks (DEQN) and Surrogates to beat the curse of dimensionality (Azinovic et al., 2022; Friedl et al., 2023).

- + **Our Innovation:** A 3-step framework: DEQN (Global Solution) → GP Surrogate → Constrained Optimization.

Outline of this talk

1. Model
2. Solution Method
 1. Deep Equilibrium Nets for IAMs
 2. Approximating Welfare Functions
 3. Finding the Optimal Tax
3. Results
4. Conclusion & Next Steps

Our Model

Integrated Assessment Models (IAMs)

- ▶ Pioneered by Nordhaus (DICE, RICE): Quantitative, often numerical.
- ▶ Key components:
 - ▶ (Simplified) climate system.
 - ▶ (Stylized) economic model of emissions AND damages.
- ▶ Economic model: needs to be dynamic, forward-looking, possibly allowing stochasticity (temperature variations, disasters, TFP), and potentially heterogeneous agents (“who is how impacted?”).

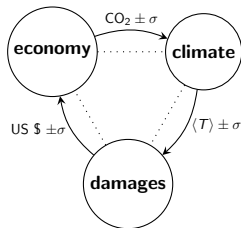


Figure: Stylized representation of an IAM.

Overview of our Stochastic IAM

- ▶ 12 Overlapping generations of selfish agents.
- ▶ 5-yearly-calibrated model: agents live from age 20 to age 80.
- ▶ Use a simple climate module (Dietz et al., 2021).
- ▶ Use a damage function with stochastic tipping (Kotlikoff et al., 2021b).
- ▶ Exogenously declining carbon intensity as in DICE-2016 (Nordhaus, 2017a).

Technology

Firms, in each period, t , produce final output, Y_t , and emissions e_t with capital, K_t and labor, L_t , and choose how much of its emissions to mitigate μ_t according to

$$(Y_t, e_t) = (\Omega_t(T_t)\Phi(\mu_t)K_t^\alpha L_t^{1-\alpha} + (1 - \delta)K_t, \kappa_t(1 - \mu_t)K_t^\alpha L_t^{1-\alpha}) \quad (1)$$

- ▶ $\Omega_t(T_t)$: climate damages that reduce output.
- ▶ $\Phi(\mu_t)$ is a mitigation cost function.
- ▶ α is the capital share.
- ▶ κ_t is carbon intensity. We assume that κ_t decreases stochastically over time as in Kotlikoff et al. (2021b).
- ▶ The abatement technology (adapted from Nordhaus, 2017b): $\Phi(\mu_t) = (1 - \theta_1\mu_t^{\theta_2})$.
- ▶ θ_1 and θ_2 are parameters and $\mu_t \in [0, 1]$.

Profit maximization is given by:

$$\begin{aligned} \max_{\mu_t, K_t, L_t} \quad & \Omega_t(T_t)(1 - \theta_1\mu_t^{\theta_2})K_t^\alpha L_t^{1-\alpha} + (1 - \delta)K_t - w_t L_t - (1 + r_t)K_t - \tau_t(1 - \mu_t)\kappa_t K_t^\alpha L_t^{1-\alpha} \\ \text{s.t.} \quad & 0 \leq \mu_t \leq 1. \end{aligned} \quad (2)$$

Households

- ▶ Agents enter the labor force at age 20 with 0 assets and die at age 80, after 12 periods.
- ▶ Agents born in period t maximize lifetime utility:

$$\mathbb{E}_t \sum_{j=1}^{12} \beta^j \frac{C_{t+j-1,j}^{1-\sigma}}{1-\sigma}, \quad (3)$$

$$\text{subject to } C_{t,j} + a_{t+1,j+1} = (1 + r_t)a_{t,j} + w_t l_j + \mathbb{T}_{t,j}, \quad (4)$$

where $C_{t,j}$, $a_{t,j}$, l_j and $\mathbb{T}_{t,j}$ reference, respectively, **consumption**, **asset level**, **labor endowment**, and **transfer** of those born at time t who are currently age j .

- ▶ l_j is similarly calibrated as in Kotlikoff et al. (2021a).
- ▶ Agents work for 8 periods, earning labor income, and then retire. Agents aged 9 to 12 work on a 40% basis as a proxy for a state pension system.
- ▶ β is the discount factor and σ is the coefficient of relative risk aversion.
- ▶ Welfare criterion is

$$U_t = \mathbb{E}_0 \sum_{j=1}^{12} \beta^j \frac{C_{t+j-1,j}^{1-\sigma}}{1-\sigma}$$

Use $\tilde{U}_t$ for welfare under taxation.

Market Clearing

- ▶ Total household assets comprise physical capital:

$$\sum_{j=1}^A a_{t,j} = K_t \quad \forall t. \quad (5)$$

- ▶ Aggregate labor endowment is the sum of individual endowments:

$$\sum_{j=1}^A l_j = L_t \quad \forall t. \quad (6)$$

- ▶ Tax income is redistributed to households each period:

$$\sum_{j=1}^A \mathbb{T}_{t,j} = \tau_t e_t \quad \forall t. \quad (7)$$

Redistribution of Tax Income

- ▶ Each period's tax income is redistributed to currently alive agents.
- ▶ In general, **transfer schemes may create winners and losers** (Kotlikoff et al., 2021a).
- ▶ Today: **2 examples of transfer policies**.
 - ▶ **"10% scheme"**: The youngest generation receives the highest share of total tax income, with each older generation getting 10% less than the next younger generation. Formally: $\mathbb{T}_{t,j} = 0.9^{(j-1)} \cdot 0.1394 \cdot \tau_t e_t$, $j \in \{2, \dots, 12\}$.
 - ▶ **Optimal transfers**: Optimal transfer shares to each generation alive in a given period (fixed at $t=0$).
- ▶ Normalization: Shares sum up to 1.

Cumulative Carbon Emissions and Temperature

Overview: Climate science shows a near-linear link between cumulative CO₂ emissions and global temperature rise:

- ▶ Global mean temperature increase (T_t^{AT}) is proportional to cumulative CO₂ emissions (E_s).
- ▶ The proportionality constant σ_{CCR} , known as the **Carbon-Climate Response (CCR)** or **TCRE**, ranges from 1.0 °C to 2.3 °C per 1 000 GtC (0.27 °C to 0.63 °C per 1 000 GtCO₂).

Equation:

$$T_t^{AT} \approx \sigma_{CCR} \sum_{s=0}^t E_s \quad (8)$$

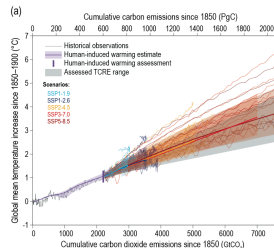


Figure: Figure from IPCC 6.

Implications:

- ▶ Equation (8) is efficient and widely used in integrated assessment models (IAMs).
- ▶ It does not account for uncertainties in σ_{CCR} nor for nonlinear effects such as potential tipping points (cf. Dietz and Venmans, 2019).

Damage Function

- ▶ Damages reduce output.
- ▶ We include stochastic climate tipping as in Weitzman (2012); Kotlikoff et al. (2021b). Climate tipping is the occurrence of climate events that lead to irreversible damage to the environment (e.g., Lenton et al., 2008).
- ▶ The damage function takes the following form:

$$\Omega_t(T_t) = \frac{1}{1 + \left(\frac{1}{\psi_1} T_t\right)^2 + \left(\frac{1}{2 \cdot TP_t} T_t\right)^{6.754}} \quad (9)$$

where ψ_1 is a damage parameter and TP_t follows a random walk which is truncated at $TP_t \in [2.5, 3.5]$.

- ▶ Once the temperature hits a tipping point, TP_t stays at the given level.

Uncertainty

- ▶ κ_t follows an AR(1) process $\kappa_t = \rho_t \kappa_{t-1} + \epsilon_{\kappa,t}$ with $\kappa_0 = 0.3502$ and $\rho_t = \rho_0 + (\rho_\infty - \rho_0)(1 - \exp(-\delta^\rho 5t))$.
- ▶ $\epsilon_{\kappa,t}$ is discretized as:

$$\epsilon_{\kappa,t} = \begin{cases} 0.03 & \text{with prob } 1/3, \\ 0.0 & \text{with prob } 1/3, \\ -0.03 & \text{with prob } 1/3. \end{cases} \quad (10)$$

- ▶ TP_t follows a random walk $TP_t = TP_{t-1} + \epsilon_{TP,t}$ with $TP_0 = 3.0$, $\epsilon_{TP,t}$ is discretized as:

$$\epsilon_{TP,t} = \begin{cases} 0.1 & \text{with prob } 1/3, \\ 0 & \text{with prob } 1/3, \\ -0.1 & \text{with prob } 1/3. \end{cases} \quad (11)$$

Equilibrium Conditions

Firm:

$$0 = \tau_t \kappa_t - \Omega_t(T_{AT,t}) \theta_1 \theta_2 \mu_t^{\theta_2 - 1} \quad (\mu_t)$$

$$r_t = (\Omega_t(T_{AT,t})(1 - \theta_1 \mu_t^{\theta_2}) - \tau_t \kappa_t (1 - \mu_t)) \alpha K_t^{\alpha - 1} L_t^{1 - \alpha} - \delta \quad (K_t)$$

$$w_t = (\Omega_t(T_{AT,t})(1 - \theta_1 \mu_t^{\theta_2}) - \tau_t \kappa_t (1 - \mu_t)) (1 - \alpha) K_t^\alpha L_t^{-\alpha} \quad (L_t)$$

Households:

For all households $j \in 1, \dots, A - 1$:

$$c_t^{-\sigma} = \beta \mathbb{E}_t \left[(1 + r_{t+1}) c_{t+1,j+1}^{-\sigma} \right] \quad (\text{EE})$$

$$C_{t,j} + a_{t+1,j+1} = (1 + r_t) a_{t,j} + w_t l_{t,j} + \mathbb{T}_{t,j} \quad (\text{BC})$$

and initial/final conditions $a_{t,1} = 0$, $a_{t+1,A} = 0$.

Market Clearing:

$$K_t = \sum_{j=1}^A a_{t,j} \quad (K_t)$$

$$L_t = \sum_{j=1}^A l_j \quad (L_t)$$

$$\tau_t e_t = \sum_{j=1}^A \mathbb{T}_{t,j} \quad (\mathbb{T})$$

The Planner's Problem

- **Objective:** The planner chooses policy parameters ϑ (taxes & transfers) to maximize the Social Welfare Function $\mathcal{W}$:

$$\begin{aligned} \max_{\vartheta} \quad & \mathcal{W}(\vartheta) = \sum_{t=-10}^{30} \gamma_t \tilde{U}_t(\vartheta) \\ \text{s.t.} \quad & \tilde{U}_t(\vartheta) \geq U_t \quad \forall t \quad (\text{if Pareto constrained}) \end{aligned} \tag{12}$$

(Where $\tilde{U}_t(\vartheta)$ is the ex-ante expected lifetime utility of generation t)

- **Welfare Metric:** Gains are reported as the consumption equivalence factor (CE_t):

$$\text{CE}_t = \left(\frac{\tilde{U}_t}{U_t} \right)^{\frac{1}{1-\sigma}} - 1 \quad (\text{where } U_t \text{ is BAU utility})$$

- **Scope & Challenge:**

- Horizon: 40 generations (11 current, 29 future).
- Complexity: Solving for the optimal high-dimensional vector ϑ in a stochastic setting.

Solution method

The Methodology: A New “Recipe”



Figure: Combining **Climate Science**, **Economic** theory, and **Artificial Intelligence** to solve complex policy problems.

Our Novel Approach in a Nutshell

► Step 1: Global solution with DEQN

- Solve the model as a function of states and a *parameterized* tax function in one go.
- For now, linear rules, e.g. in cumulative emissions, κ , TP :
→ 4 pseudo-states $\vartheta = (\vartheta_0, \vartheta_E, \vartheta_\kappa, \vartheta_{TP})$, plus 11 from transfers.

⇒ Once trained, the solution to the problem allows us to simulate data essentially for free.

► Step 2: Surrogate for welfare

- Build a surrogate to approximate aggregate welfare as a function of the parameters of the tax function using Gaussian Process Regression & Bayesian Active Learning.¹

► Step 3: Optimize the planner's objective

- Optimize planner objective: maximize the weighted sum given a specific set of welfare weights.

Companion notebook (this lecture). Notebook 09 demonstrates the Step-2 machinery – a Gaussian-process surrogate (Matérn-5/2 + white noise) with **Sobol** sensitivity analysis – applied to the social cost of carbon of the *deterministic DICE* model over (ρ, π_2) . The full stochastic-OLG / DEQN solve (Step 1) and the constrained Pareto-tax optimization (Step 3) are research results, not reproduced by the notebook.

¹Surrogate is a 21st century lookup table: $\text{Welfare}(\vartheta_0, \dots, \vartheta_{15})$

Recall: Steps of the Method (2 Surrogates)

Step 1: DEQN

$$P(x_{state,t}, \vartheta)$$

Step 2: GP(θ)

$$\widehat{\text{QoI}}(\vartheta)$$

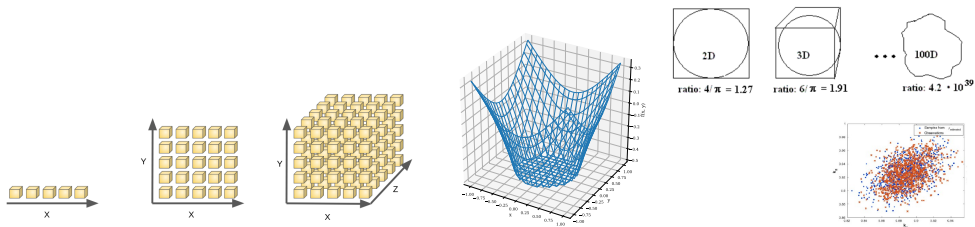
Step 3: Optimizer

$$\arg \max \widehat{\text{QoI}}(\vartheta)$$

Deep Equilibrium Nets for IAMs

Solving Stochastic IAMs

1. Models suffer from the curse of dimensionality.
2. Models suffer from non-linearities.
3. Have to approximate and interpolate high-dimensional functions on irregular-shaped geometries.



→ “Deep Equilibrium Nets” (Azinovic et al., 2022).

Deep equilibrium nets (Azinovic et al., 2022)

A **functional rational expectations equilibrium**: $\{f_i\}_{i=1}^{N_{\text{out}}}$, where

$$f_i : \mathcal{D} \subset \mathbb{R}^{N_{\text{in}}} \rightarrow \mathbb{R} : \underbrace{\mathbf{x}}_{\text{state}} \rightarrow \underbrace{f_i(\mathbf{x})}_{\text{endogenous variables}}, \text{ s.t. : } \underbrace{\mathbf{G}(\mathbf{x}, f_1, \dots, f_{N_{\text{out}}}) = 0}_{\text{equilibrium conditions}}$$

A **deep equilibrium net**: $\mathcal{N}_\rho$, where

$$\mathcal{N}_\rho : \mathcal{D} \subset \mathbb{R}^{N_{\text{in}}} \rightarrow \mathbb{R}^{N_{\text{out}}} : \underbrace{\mathbf{x}}_{\text{state}} \rightarrow \underbrace{\mathcal{N}_\rho(\mathbf{x})}_{\text{approximate endogenous variables}} \approx \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_{N_{\text{out}}}(\mathbf{x}) \end{bmatrix}$$

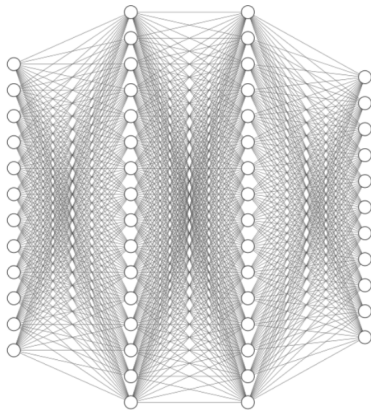
Preview of key ideas:

1. Use the definition of the equilibrium functions, *i.e.* the implied error in the optimality conditions, as loss function.
2. Learn the equilibrium functions with stochastic gradient descent.
3. Take the data points from a simulated path.

What is a deep neural net?

input $:= \mathbf{x} \rightarrow \phi^1(W_\rho^1 \mathbf{x} + \mathbf{b}_\rho^1) =: \mathbf{hidden\ 1}$
 $\rightarrow \mathbf{hidden\ 1} \rightarrow \phi^2(W_\rho^2(\mathbf{hidden\ 1}) + \mathbf{b}_\rho^2) =: \mathbf{hidden\ 2}$
 $\rightarrow \mathbf{hidden\ 2} \rightarrow \phi^3(W_\rho^3(\mathbf{hidden\ 2}) + \mathbf{b}_\rho^3) =: \mathbf{output}$

The neural net is then given by the choice of activation functions and the parameters ρ .



Our loss function

As a loss function, we implement

$$l_{\rho} := \frac{1}{N_{\text{path length}}} \sum_{\mathbf{x}_i \text{ on sim. path}} (\mathbf{G}(\mathbf{x}_i, \mathcal{N}_{\rho}))^2$$

where we use $\mathcal{N}_{\rho}$ to simulate a path. $\mathbf{G}$ is chosen, such that the true equilibrium policy $\mathbf{f}(\mathbf{x})$ is defined by $\mathbf{G}(\mathbf{x}, \mathbf{f}) = 0 \ \forall \mathbf{x}$. Therefore, there is **no need for labels** to evaluate our loss function.

Training Deep Equilibrium Nets²

1. Simulate a sequence of states $\mathcal{D}_{\text{train}}^i \leftarrow \{\mathbf{x}_1^i, \mathbf{x}_2^i, \dots, \mathbf{x}_T^i\}$ from the policy encoded by the network parameters ρ^i .
2. Evaluate the errors of the equilibrium conditions on the newly generated set $\mathcal{D}_{\text{train}}$.
3. If the error statistics are not low enough:
 - 3.1 Update the parameters of the neural network with a gradient descent step (or a variant):

$$\rho_k^{i+1} = \rho_k^i - \alpha_{\text{learn}} \frac{\partial \ell_{\mathcal{D}_{\text{train}}^i}(\rho^i)}{\partial \rho_k^i}.$$

- 3.2 Set new starting states for simulation: $\mathbf{x}_0^{i+1} = \mathbf{x}_T^i$.
- 3.3 Increase i by one and go back to step 1.

²Sample codes here: <https://github.com/sischei/DeepEquilibriumNets>.

Mapping the OLG Model to DEQN

Goal: Approximate the policy functions of the OLG model using a neural network.

Inputs (state vector):

$$x = [t, TP_t, TP_{\text{reached}}, \kappa_t, a_{t,j}, E_t, \vartheta] \in \mathbb{R}^{A+5+d_\vartheta}$$

Outputs (policy vector):

$$y = [a_{t+1,j+1}, v_{t,j}] \in \mathbb{R}^{2(A-1)}$$

Policy map:

$$\Pi : x \Rightarrow y$$

Why include ϑ ?

- ▶ Parameters of the tax rule $\tau(\vartheta)$.
- ▶ Transfer shares included as pseudo-states.
- ▶ *DEQN learns policies for all tax rules and transfers at once.*

Performance:

- ▶ Runtime from scratch: $\approx 4\text{h}$ on laptop.
- ▶ Warm start: a few minutes.
- ▶ Accuracy: $\text{MSE} \sim 10^{-6}$, $\text{EE} \sim 10^{-3}$.

Architecture of the Deep Neural Network

Hyperparameter	Value
Optimizer (LR)	Adam (1e-5)
Batch size	64
NN hidden layers	2
NN units	256
activation function	GeLu
N inputs	$12 + 5 + \vartheta$
N outputs	$2 \cdot (A - 1) = 22$
N epochs per episode	1
N sim episode length	70 (each step is 5y)
N_batches	512

Table: Hyperparameters of neural network.

Approximating Welfare Functions

Approximating Welfare with Surrogates

The Computational Bottleneck:

- ▶ DEQN solves the equilibrium, but evaluating expected welfare $\mathbb{E}[\mathcal{W}]$ requires expensive Monte Carlo simulations (10,000 paths per policy ϑ).
- ▶ **Solution:** Train a **Gaussian Process (GP)** to map policy parameters ϑ directly to welfare outcomes.

Two Optimization Targets (Qols):

1. Maximize Aggregate Welfare

Goal: Maximize scalar $\mathcal{W}$.

$$\max_{\vartheta} \mathbb{E}_0[\mathcal{W}(\vartheta)]$$

- ▶ **Surrogate:** Single scalar GP.
- ▶ **Implication:** Ignores distribution (allows losers).

2. Pareto Improvement

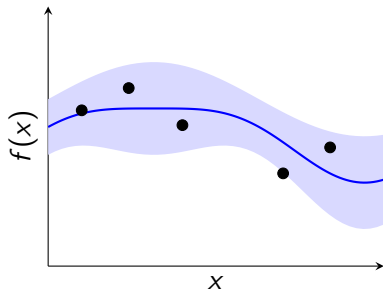
Goal: Maximize $\mathcal{W}$ subject to:

$$\tilde{U}_t(\vartheta) \geq U_t \quad \forall t$$

- ▶ **Surrogate:** Vector of 40 GPs (one per cohort).
- ▶ **Implication:** No generation is worse off.

The Engine: Gaussian Processes & Active Learning

1. Gaussian Processes (GP)



Kernel:

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right)$$

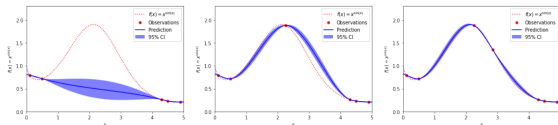
Predictive mean & variance:

$$\mu(x^*) = k_*^\top (K + \sigma_n^2 I)^{-1} \mathbf{y}$$

$$\sigma^2(x^*) = k(x^*, x^*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_*$$

2. Bayesian Active Learning (BAL)

$$\alpha(x) = \underbrace{\mu(x)}_{\text{Exploit}} + \kappa \cdot \underbrace{\sigma(x)}_{\text{Explore}}$$



We simulate points that maximize this score, reducing computational cost by orders of magnitude.

Constructing the Welfare Surrogate

1. The Mapping

We map policy rules directly to **Expected Welfare**:

$$\vartheta \longmapsto \mathbb{E}_0[\mathcal{W}(\vartheta)]$$

- **Input (ϑ):** Tax and transfer parameters.
- **Target:** Monte Carlo mean of lifetime utilities.

2. Active Training

We populate the training set $\mathcal{D}$ efficiently:

- **Initial Design:** 450 points sampled randomly (Latin Hypercube).
- **Refinement:** 50 points added via [Bayesian Active Learning \(BAL\)](#).
- **Total:** Only 500 simulations needed.

3. The Model

Fitting the Gaussian Process:

- **Kernel:** Squared Exponential with Automatic Relevance Detection (ARD).
- **Accuracy:** Leave-One-Out (LOO) error $\approx 10^{-4}$.
- **Result:** A smooth, cheap-to-query surface.

Finding the Optimal Tax

Maximizing Welfare

- ▶ We now have a cheap-to-evaluate surrogate model to map **tax functions onto welfare**. But we still need to find the optimal tax.
- ▶ We aim to **find the arg max of our surrogate model**, which gives us the optimal tax function parameters:

$$\vartheta^* = \arg \max_{\vartheta} \widehat{\text{QoI}}(\vartheta)$$

s.t. various constraints.

- ▶ We use **constrained optimization with linear constraints** for periods $\{0, T\}$ that ensure that the parameters of the tax function stay within the feasible range during optimization.

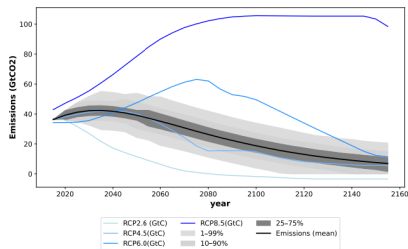
Results

Calibration

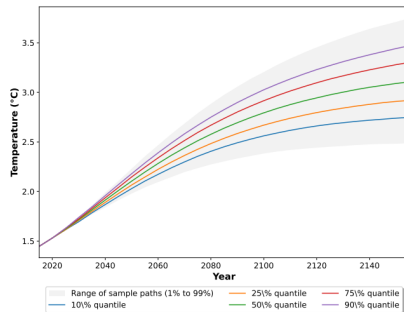
Symbol	Parameter	Value
β	Discount factor	0.9
σ	Relative risk aversion	3
α	Capital share	0.3
δ	Depreciation	0.2
σ_{ccr}	Climate sensitivity	1.7
ψ_1	parameter damage function	13.16
θ_1	Intercept of control cost function	0.7
θ_2	Exponent of control cost function	2.6
κ_0	Initial carbon intensity	0.35032
ρ_0	Initial AR(1) param for carbon intensity	1.08
ρ_∞	Final AR(1) param for carbon intensity	0.91
δ^ρ	AR(1) param depreciation for carbon intensity	0.04
E_0	initial stock of carbon thGtC	0.851
L	Labor (to match initial emissions)	600

Table: Parameterization of the OLG Model.

BAU Scenario versus RCP Scenarios



(a) Emissions



(b) Temperature

Left: Global CO₂ Emissions

- ▶ Comparison of our model's BAU against IPCC Representative Concentration Pathways (RCP).
- ▶ Our endogenous emissions path tracks closely with **RCP 4.5**.
- ▶ Base year (0) corresponds to 2017.

Right: Global Warming Projections

- ▶ Projected mean warming reaches $\approx 3^{\circ}\text{C}$ over 150 years.
- ▶ Substantial uncertainty exists: range of **2.5°C to 3.6°C** .
- ▶ Highlights significant tail risk in the absence of policy.

Three Policy Scenarios

► **Scenario 1: Unconstrained Welfare Maximization**

Linear tax on cumulative emissions + exogenous transfers.

$$\tau_t(E_t) = \vartheta_0 + \vartheta_E \cdot E_t$$

► **Scenario 2: Pareto-Improving Policy**

Linear tax on cumulative emissions + **optimal** transfers.

$$\tau_t(E_t) = \vartheta_0 + \vartheta_E \cdot E_t$$

► **Scenario 3: Increasing Policy Complexity**

Linear tax on cumulative emissions, carbon intensity (κ), and tipping risk (TP) + optimal transfers.

$$\tau_t(\cdot) = \vartheta_0 + \vartheta_E E_t + \vartheta_\kappa \kappa_t + \vartheta_{TP}(1 - \mathbb{D}_{TP})$$

Scenario 1: Unconstrained Welfare Maximization

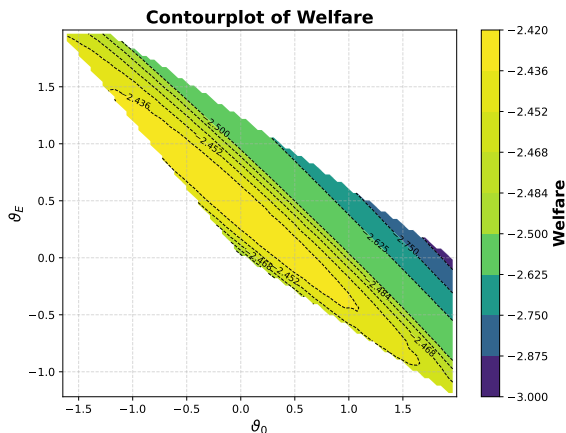
The youngest generation receives the largest share of total tax revenue, and the share for each older generation decreases by 10% relative to the next younger one. The shares are normalized to sum to one. Formally, our transfer scheme, $\mathbb{T}$, is defined as:

$$\mathbb{T}_{t,j} = 0.9^{(j-1)} \cdot 0.1394 \cdot \tau_t e_t, \quad j \in \{2, \dots, 12\}. \quad (13)$$

The resulting constrained optimization problem reads as:

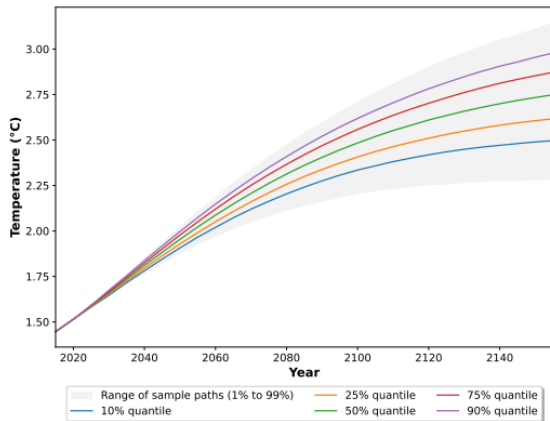
$$\begin{aligned} \vartheta^* &= \arg \max_{\vartheta \in [a,b]} \widehat{\text{QoI}}(\vartheta) \\ &\text{s.t.} \\ 0 &\leq \vartheta_0 + \vartheta_E E_0 \leq \bar{\tau}, \\ 0 &\leq \vartheta_0 + \vartheta_E E_{29} \leq \bar{\tau}. \end{aligned} \quad (14)$$

The Welfare Surrogate (Scenario 1)



- **The Surface:** A Gaussian Process approximation of the Social Welfare Function.
- **The Axes:** Tax Intercept (ϑ_0) vs. Slope (ϑ_E).
- **The Goal:** Find the "peak" (yellow zone) without running millions of simulations.
- **Precision:** Leave-One-Out error $\approx 10^{-5}$.

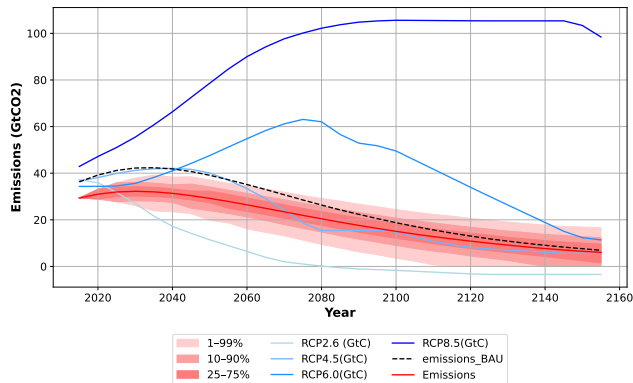
Temperature (Scenario 1: Unconstrained)



Outcomes:

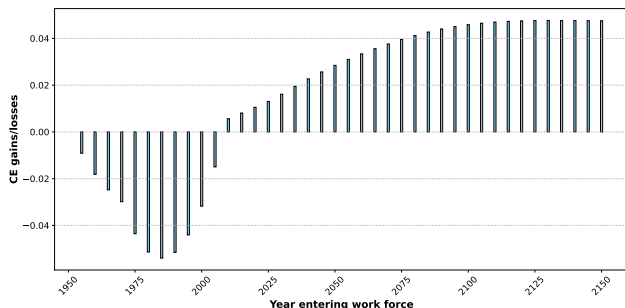
- ▶ **Aggressive Mitigation:** Stabilizes mean warming at ≈ 2.7 °C.
- ▶ **Tail-Risk Insurance:** The 99th percentile is capped at ≈ 3.1 °C.
- ▶ **The Mechanism:** Requires immediate, heavy taxation ($\approx 70\%$ average after 150 years).
- ▶ **The Cost:** Imposes up to **-5% welfare losses** on current older cohorts.

Emissions (Welfare Maximizing)



- **Immediate Shock:** Emissions drop $\approx 25\%$ below BAU within the first 50 years.
- **Eliminating the Extremes:** The policy successfully truncates the high-emission tail (the red shaded area).
- **Convergence:** Post-2100, the gap narrows as exogenous carbon intensity declines naturally.

Optimal Welfare Distribution



The Price:

- **Aggregate Welfare Gain:** The total pie grows by **1.6%** (consumption equivalent).
- **Future Gain:** Generations born after 2100 enjoy gains of $\approx 4.8\%$.
- **Current Pain:** This comes at a steep cost. Transition generations suffer losses up to **5%**.
- **Conclusion:** Without transfers, this optimal policy is politically dead on arrival.

The Challenge

- ▶ The welfare-maximizing tax (simple example) reduces warming to 2.7°C.
- ▶ **But:** It imposes welfare losses of up to **5%** on current older generations.
- ▶ **Result:** The policy is economically efficient but politically infeasible.
- ▶ **Goal:** Can we design a tax *and* transfer scheme where $\tilde{U}_t \geq U_t$ for all t ?

Scenario 2: Pareto-Improving Policy

For 40 values of t :

$$\mu_{*,t}(\vartheta) = \widehat{QoI}_t(\vartheta) = \mathbb{E}[\tilde{U}_t(\vartheta)] \quad \forall t \in \{-10, \dots, 29\} \quad (15)$$

Solve:

$$\begin{aligned} \vartheta^* &= \arg \max_{\vartheta \in [a,b]} \sum_{t=-10}^{29} \gamma_t \widehat{QoI}_t(\vartheta) = \arg \max_{\vartheta \in [a,b]} \sum_{t=-10}^{29} \gamma_t \mathbb{E}_0[\tilde{U}_t(\vartheta)] \\ \text{s.t.} \\ \mathbb{E}_0[\tilde{U}_t(\vartheta)] &\geq \mathbb{E}_0[U_t] \quad \forall t, \\ 0 &\leq \vartheta_0 + \vartheta_E E_0 \leq \bar{\tau}, \\ 0 &\leq \vartheta_0 + \vartheta_E E_{29} \leq \bar{\tau}, \\ \sum_{i=1}^{12} \vartheta_i &= 1, \end{aligned} \quad (16)$$

Optimal Parameters and Transfers

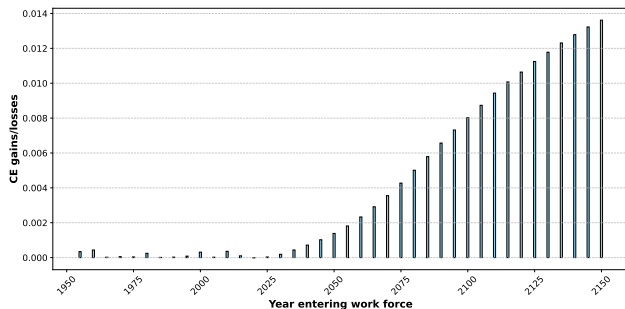
We find optimal tax parameters: $\vartheta_0 = -0.186$ and $\vartheta_E = 0.225$.

The optimal transfer shares allocate resources to specific cohorts to satisfy the Pareto constraints:

t1	t2	t3	t4	t5	t6	t7	t8	t9	t10	t11	t12
0.128	0.051	0.058	0.09	0.15	0.09	0.066	0.143	0.076	0.048	0.039	0.06

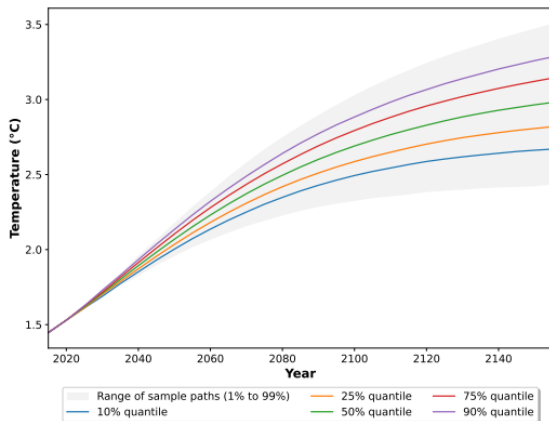
Table: Pareto optimal transfer scheme. Optimal shares for each concurrent cohort i under the linear cumulative emissions tax.

Optimal Welfare (Pareto Improving)



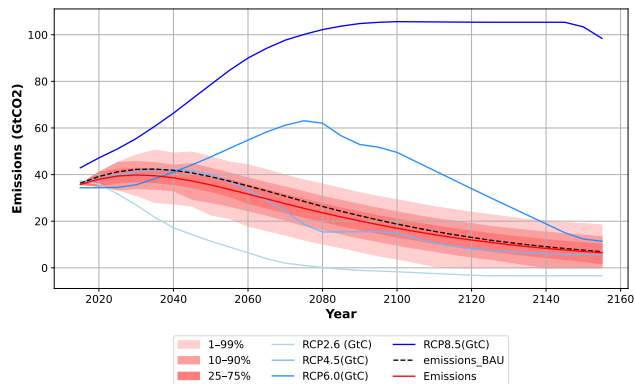
- **The Pareto Constraint:** Transfers are explicitly optimized to keep initial generations (and those born in the next decade) at **BAU levels** (0% loss).
- **Future Gains:** Welfare improves monotonically, reaching $\approx 1.4\%$ for distant generations.
- **The Cost of Consensus:** Compared to the unconstrained case (max 4.8%), future gains are smaller, but **no generation loses**.
- **Aggregate:** Total welfare gain of **0.42%**.

Temperature (Scenario 2: Pareto-Improving)



- **The Trade-off:** Mean warming settles just below **3 °C** (higher than Scenario 1).
- **Crucial Insurance:** The 99th percentile **never exceeds 3.5 °C**.
- **Tail Truncation:** The policy deliberately eliminates the catastrophic tail (>3.6 °C under BAU).
- **The Win:** Achieved with optimal transfers → **no generation loses**.

Emissions: Mitigating Risk



- **Modest Average Reduction:** The mean emissions path is not drastically lower than BAU (required to preserve current consumption).
- **Tail Truncation:** The policy's primary physical impact is eliminating the high-emissions upside.
- **Result:** The 99th percentile of emissions drops below 40 GtCO₂ by 2070 (20 years earlier than BAU).
- **Conclusion:** The Pareto-optimal policy functions primarily as disaster insurance rather than aggressive mitigation.

Scenario 3: Increasing Policy Complexity

Question: Can we improve welfare by targeting specific climate drivers?

We expand the tax rule to include **Carbon Intensity** (κ_t) and **Tipping Risk** (TP_t):

$$\tau_t = \vartheta_0 + \vartheta_E \frac{E_t}{E_0} + \underbrace{\vartheta_\kappa \frac{\kappa_t}{\kappa_0}}_{\text{Intensity}} + \underbrace{\vartheta_{TP}(1 - \mathbb{D}_{TP})}_{\text{Tipping Risk}} \quad (17)$$

- ▶ $\mathbb{D}_{TP} \in [0, 1]$: Normalized distance to the tipping threshold.
- ▶ **Dimensionality:** This adds new pseudo-states, increasing the parameter space to $d_\vartheta = 16$ (4 tax parameters + 12 transfer shares).
- ▶ We use the same ML-based optimization to find the constrained Pareto-optimal policy.

Results: Diminishing Returns to Complexity

We find the optimal parameters:

$$\{\vartheta_0, \vartheta_E, \vartheta_\kappa, \vartheta_{TP}\} = \{-0.237, 0.203, 0.037, 0.012\}$$

Key Finding: Complexity adds very little welfare value.

Policy Specification	Instruments	Welfare Gain*
Linear in Cumulative Emissions (E_t)	2	0.42%
Linear in E_t, κ_t, TP_t	4	0.45%

Table: Aggregate welfare gains in consumption-equivalent terms.

Conclusion:

A simple, well-designed tax on cumulative emissions captures **93%** of the potential welfare gains. Adding complex state-dependencies provides only marginal benefits.

The Cost of Political Feasibility

Comparing the physical outcomes of the two approaches reveals a crucial trade-off:

1. Welfare Maximizing

(Unconstrained)

- ▶ Aggressive early mitigation.
- ▶ Warming stabilizes at $\approx 2.7^{\circ}\text{C}$.
- ▶ Current generations lose (up to 5%).

2. Pareto Improving

(Constrained)

- ▶ Slower initial mitigation path.
- ▶ Warming stabilizes at $\approx 3.0^{\circ}\text{C}$.
- ▶ No generation loses.
- ▶ **Crucial: Truncates worst-case damages (insurance).**

Trade-off & Insurance

Ensuring political support requires delaying some mitigation, leading to higher *average* temperatures ($\approx +0.3^{\circ}\text{C}$). However, it successfully **eliminates extreme damage scenarios** ($> 15\%$ of GDP), effectively insuring future generations against climate disasters.

Conclusion

- **Methodological Contribution:**

A scalable 3-step ML framework (DEQN + GP + Optimization) to solve previously intractable, high-dimensional OLG-IAMs.

- **Core Policy Result:**

Simple linear tax + Optimal transfers → **Pareto Improvement.**

Yields **0.42%** aggregate welfare gain.

- **Insight on Complexity:**

Adding intensity & tipping targets yields only marginal gains (to **0.45%**). **Simple, implementable rules capture the vast majority of benefits.**

- **Broader Implications:**

Provides a generic blueprint for optimal fiscal and climate policy design in complex heterogeneous-agent environments.



Questions?

Appendix: Sampling Domain

Model	Planner Target	Parameter	$\underline{\vartheta}_i$	$\bar{\vartheta}_i$
Linear in E	Max. Welfare	ϑ_0	-2.0	2.0
	Max. Welfare	ϑ_E	-2.0	2.0
	Max. Welfare	τ	0.0	1.5
Linear in E + Transfers	Pareto	ϑ_0	-0.5	0.1
	Pareto	ϑ_E	0.0	0.5
	Pareto	τ	0.0	0.8
Full linear + Transfers	Pareto	ϑ_0	-1.0	0.5
	Pareto	ϑ_E	-0.5	1.0
	Pareto	ϑ_κ	-0.6	0.6
	Pareto	ϑ_{TP}	-1.0	1.0
	Pareto	τ	0.0	0.8

Table: Parameter bounds across models and objectives.

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